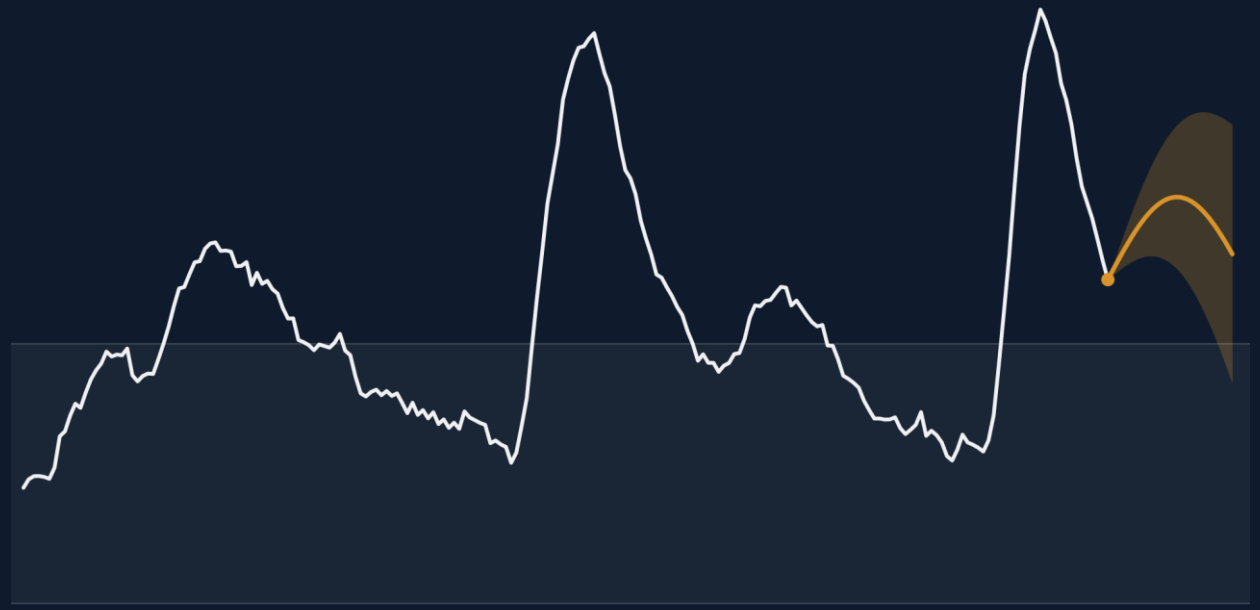
DIGITAL TWIN CHALLENGE 2026

MadhuTwin

A physiological–AI digital twin that warns of glycaemic adverse events in type 2 diabetes before they happen



A real glucose day and the twin's next two hours

Anish Karthik Ayyagari

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THE BUILDER

One builder, every layer: physiology, machine learning, data and product

Anish Karthik Ayyagari

Indian Institute of Technology Kharagpur

SOLO ENTRANT AND TEAM LEADER

Designed, built and validated end to end, from the glucose–insulin equations to the clinician dashboard. Every number in this deck is reproducible from open or synthetic data.

01 **Physiological twin**

Glucose–insulin model with Indian meals and medicines

02 **Personalisation**

Health-record prior × sensor likelihood, MAP fit

03 **Hybrid ML and validation**

Leave-patients-out study on real CGMacros data

04 **Bharat synthetic cohort**

80 fictional Indian patients, full EHR and wearables

05 **API and copilot**

FastAPI service, FHIR export, grounded summaries

06 **Clinician dashboard**

Triage, live twin, what-if, AGP report, evidence

India has 101 million people with diabetes, yet their glucose is seen a few times a year

101 M
adults living with diabetes

136 M
adults with prediabetes

11.4 %
national diabetes prevalence

15.3 %
prediabetes prevalence

ICMR-INDIAB national study, Anjana et al., Lancet Diabetes & Endocrinology 2023 (113,043 adults, 31 states and union territories).

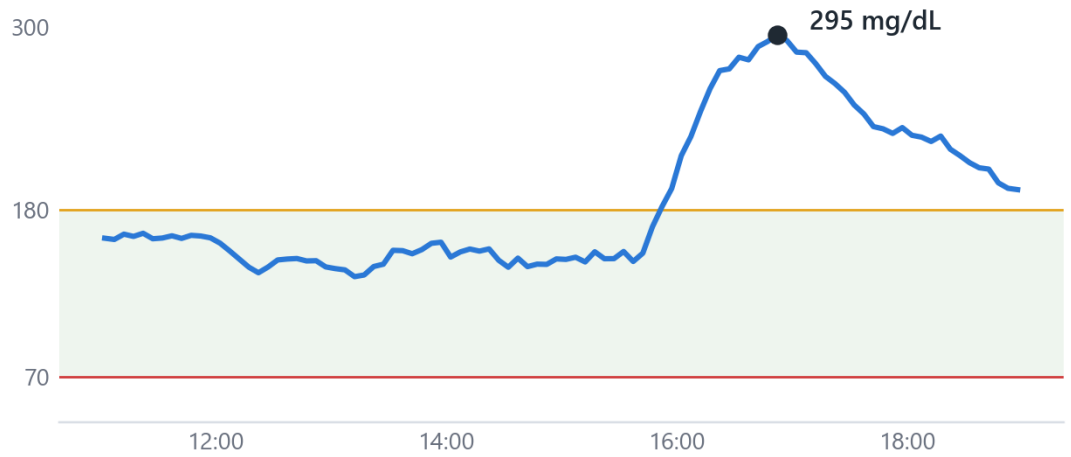
Care is reactive. A quarterly HbA1c averages three months into one number. It cannot show the post-meal spikes or the afternoon lows behind complications and emergencies.

Sensors arrive; decisions do not. CGMs and smartwatches now produce thousands of readings per patient, but doctors receive data dumps, not early warnings.

Two adverse events slip through between clinic visits

POST-MEAL SPIKE

High-glycaemic-load meals push glucose past 250 mg/dL for hours

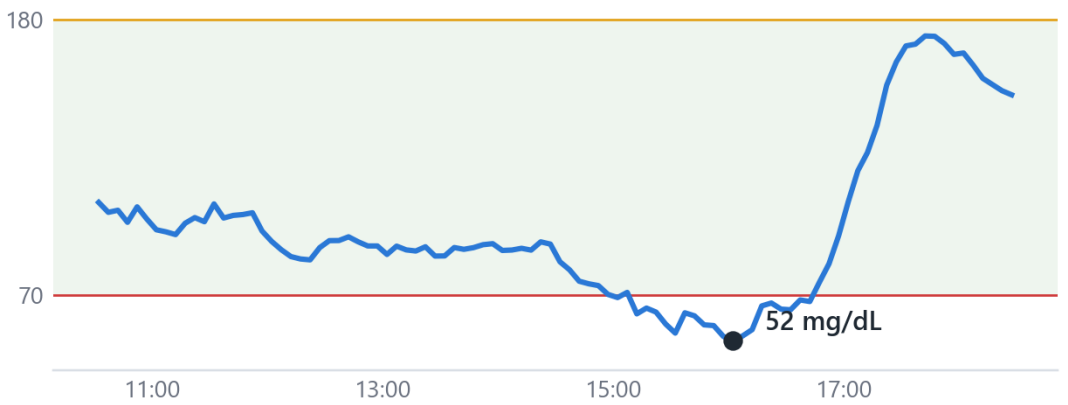


Sourav, 45, Kolkata · late lunch · synthetic cohort

The same meal raises glucose very differently from one person to the next.

DRUG-INDUCED LOW

A sulfonylurea plus a skipped meal or a long walk pushes glucose below 70



Ramesh, 61, Mysuru · glimepiride and CKD · synthetic cohort

Inexpensive and widely prescribed in India, sulfonylureas are a leading cause of severe lows.

A living model of each patient, fed by their health record and their sensors

WHAT GOES IN

Static record

Demographics and diagnoses

HbA1c, fasting glucose and insulin, lipids, eGFR

Medications, TCF7L2, polygenic risk

Live streams

CGM every 5 minutes

Heart rate, steps, METs

Sleep stages, HRV, meal log

THE TWIN

Personalised physiology, corrected by machine learning

Glucose–insulin model with Indian meals and medicines

Personalised from the record, refined by sensor data

Synced to every CGM reading

ML corrects the residual, with calibrated uncertainty and reasons

WHAT COMES OUT

Forecast

Two hours ahead, with 90 % intervals

Early warning

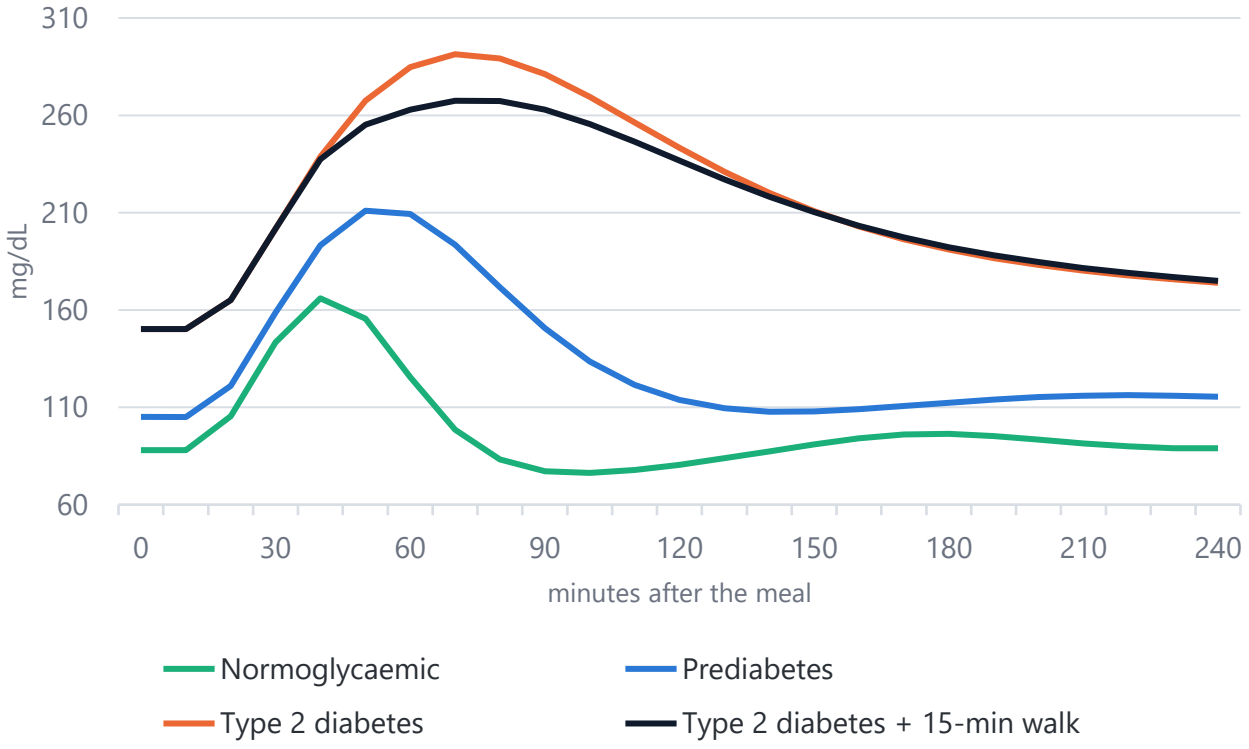
Hyperglycaemia, spikes and lows, with reasons

What-if

Meals, walks, sleep and drug changes, simulated

Physiology first: a glucose–insulin model that understands Indian meals and medicines

Same rice-and-sambar lunch (85 g carbohydrate), simulated for four physiologies



1 Meal absorption

Fast and slow chains: glycaemic index, fat, fibre and protein shape the curve

2 Insulin secretion and action

Beta-cell response, insulin sensitivity, glucose effectiveness

3 Activity, sleep, rhythm

METs raise uptake; short sleep lowers sensitivity; dawn and circadian effects

4 Medicines used in India

Metformin, sulfonylureas, DPP-4i (teneligliptin), SGLT2i, voglibose, pioglitazone, basal insulin

The health record sets the prior; the sensors make it personal

1

Prior from routine labs

HbA1c, fasting glucose and insulin (HOMA-IR, HOMA-B) and weight set the patient's starting physiology, so the twin is useful on day zero.

2

Onboarding fit

The first days of CGM, meals and activity refine the parameters by fitting the twin's own two-hour forecasts (maximum a posteriori).

3

Continuous sync

Every CGM sample corrects the virtual patient; insulin, gut glucose and exercise effects evolve through the physiology.

41.9 → 29.7

mg/dL two-hour error of the physiology alone, before and after onboarding (real patients)

ρ = −0.68

twin insulin sensitivity vs lab HOMA-IR across 45 real participants

An interpretable phenotype. Insulin sensitivity, beta-cell response, disposition index, absorption speed, exercise response and dawn effect, normally measured with a clamp study, estimated from data the patient already generates.

Machine learning corrects physiology, and says how sure it is

1 Residual forecaster

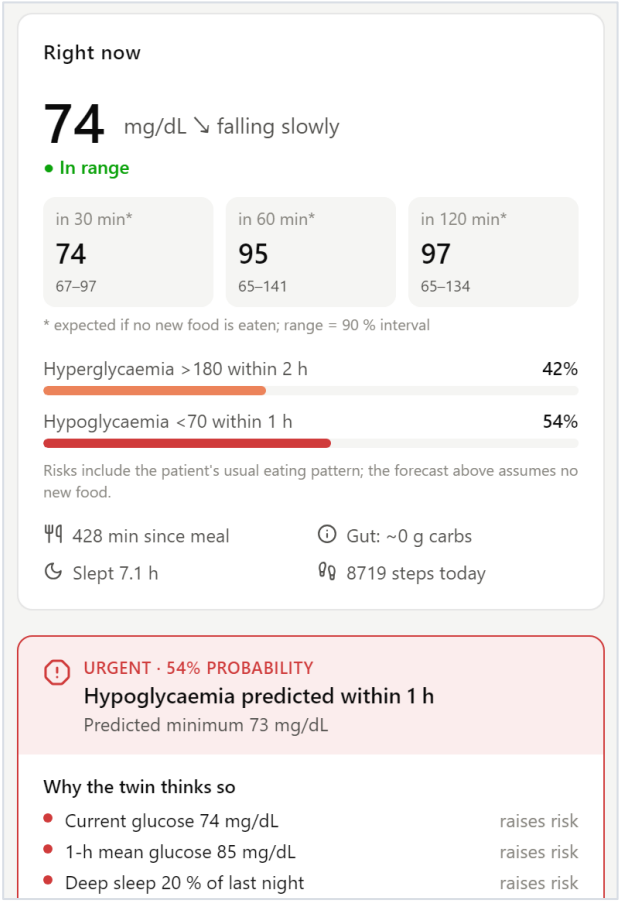
Gradient-boosted trees for each horizon (30–120 min) see every stream plus the twin's own forecast and learn when to trust physiology. Forecasts assume no new food; new meals enter through the twin.

2 Conformal intervals

Conformalised quantile regression gives 90 % prediction intervals with a distribution-free guarantee. Observed coverage on unseen patients: 91.7 % at two hours.

3 Explained alerts

Calibrated risks for hyperglycaemia, spikes and hypoglycaemia. Exact TreeSHAP values, grouped by data stream, tell the doctor why each alert fired.



Live view: risk, reasons and evidence by stream

Ramesh's afternoon low, flagged before it happens



Pale blue after 'Now' is what actually happened, revealed for the demo. Synthetic patient.

1 Who

61, glimepiride and CKD; skips lunch and walks to the market

2 What the twin sees

A falling trend, recent activity, a long fast and a sulfonylurea

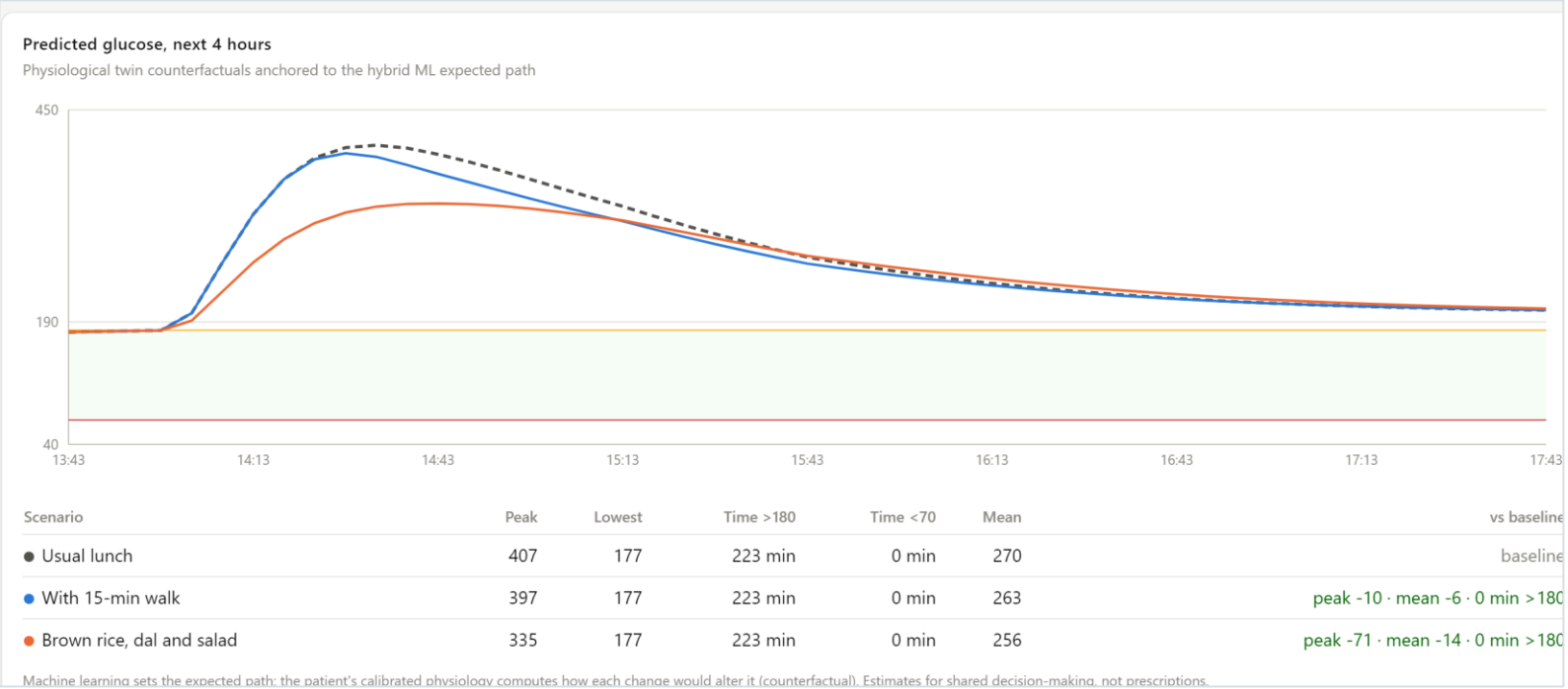
3 Risk, not just a line

The median forecast expects recovery; the calibrated risk model gives a 54 % chance of a low. The low happens.

4 What the doctor gets

An urgent alert with reasons, a twin-tested action (15 g carbohydrate) and a prompt to review the dose

What-if: test a meal swap or a walk on the patient's own physiology



Scenario	Peak	Lowest	Time >180	Time <70	Mean	vs baseline
Usual lunch	407	177	223 min	0 min	270	baseline
With 15-min walk	397	177	223 min	0 min	263	peak -10 · mean -6 · 0 min >180
Brown rice, dal and salad	335	177	223 min	0 min	256	peak -71 · mean -14 · 0 min >180

Machine learning sets the expected path; the patient's calibrated physiology computes how each change would alter it (counterfactual). Estimates for shared decision-making, not prescriptions.

Lakshmi, 54, Chennai: usual rice-and-rasam lunch, the same lunch with a 15-minute walk, and brown rice with dal. Twin estimates.

−71 mg/dL

peak with brown rice and dal instead of rice and rasam

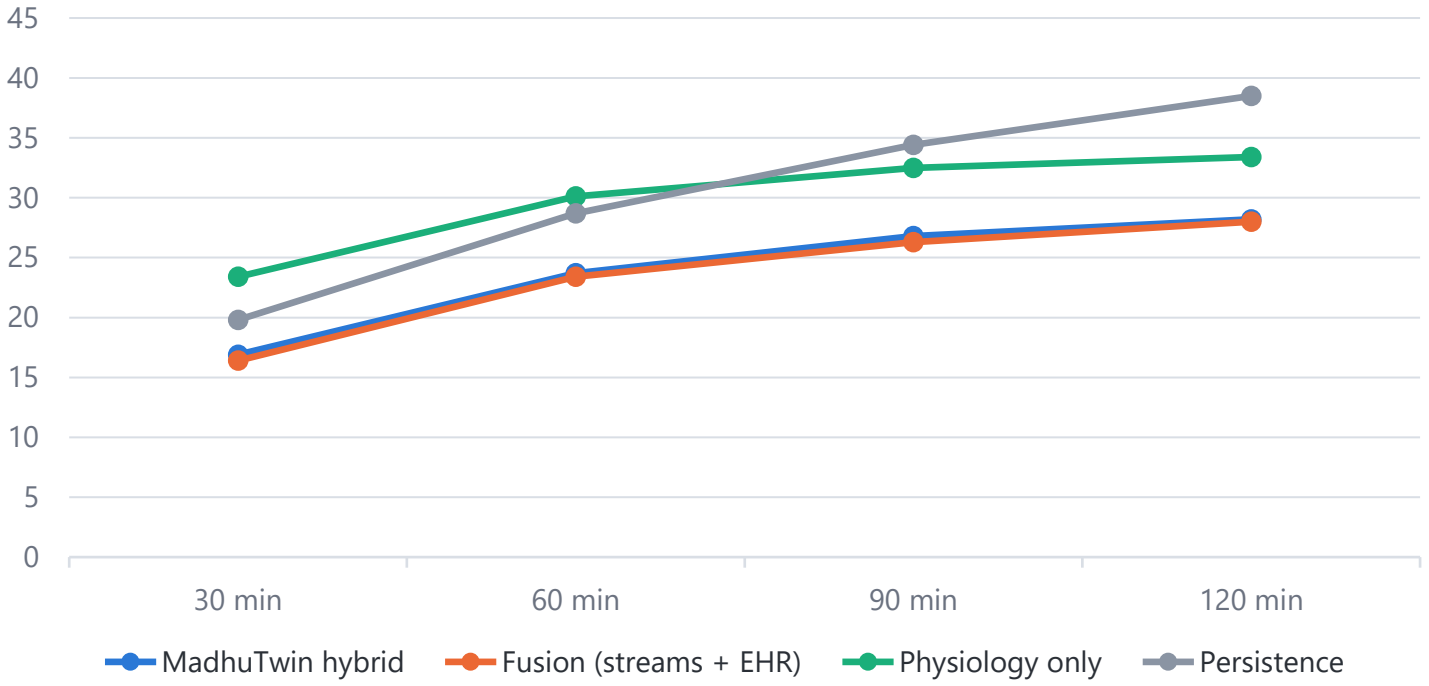
−10 mg/dL

peak with a 15-minute walk after the usual lunch

Machine learning sets the expected path; the patient's calibrated physiology computes how each change alters it.

On unseen real patients, two-hour error is 27 % below persistence

Forecast RMSE on unseen CGMacros patients, mg/dL (lower is better)



45 adults with CGM, Fitbit, meals and labs · 5-fold leave-patients-out · twins personalised only on each patient's first days · 311 unseen patient-days · native sensor samples only

28.2 mg/dL

two-hour RMSE, vs 38.5 for persistence

99.0 %

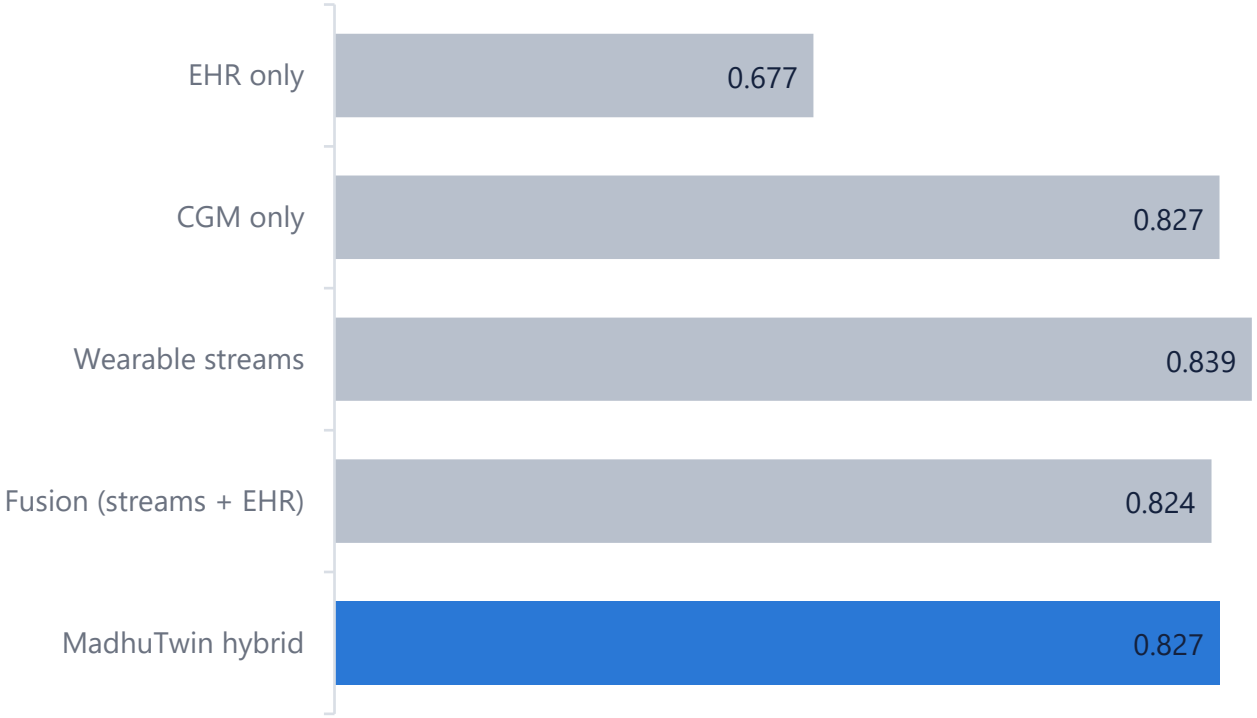
of two-hour forecasts clinically acceptable (Clarke zones A+B)

91.7 %

observed coverage of the 90 % interval

The record tells you who is at risk; the live streams tell you when

Predicting hyperglycaemia onset within two hours: AUROC by inputs



75 min

median warning before glucose crosses 180 mg/dL

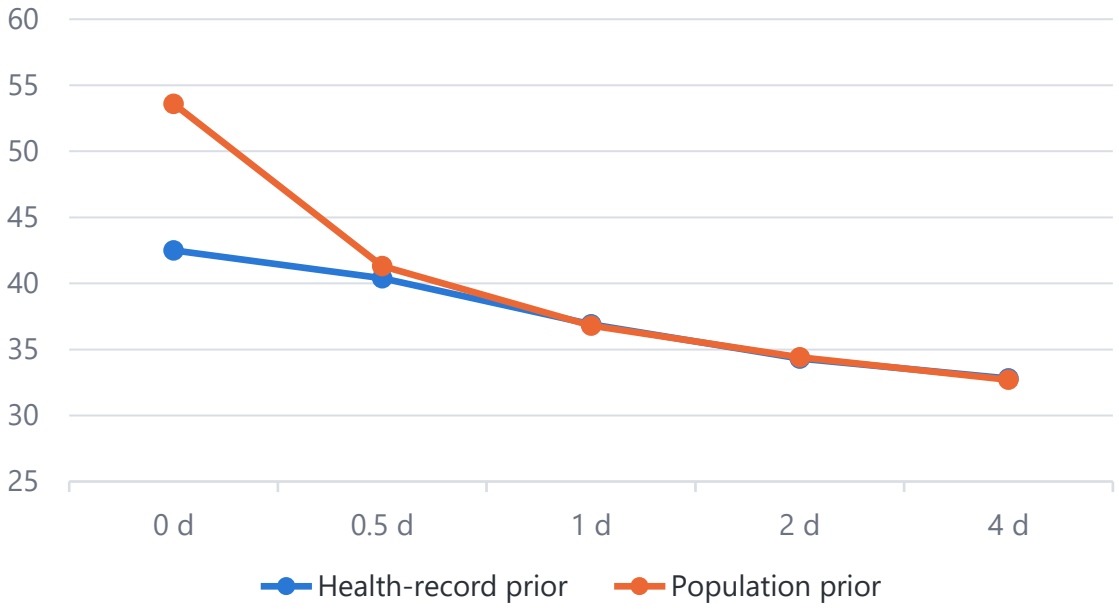
61 %

of hyperglycaemic episodes flagged at least 15 minutes ahead

Record alone: AUROC 0.68 (who is at risk). With live streams: 0.83 (when). Spike AUROC 0.82; 1.4 false alerts per patient-day.

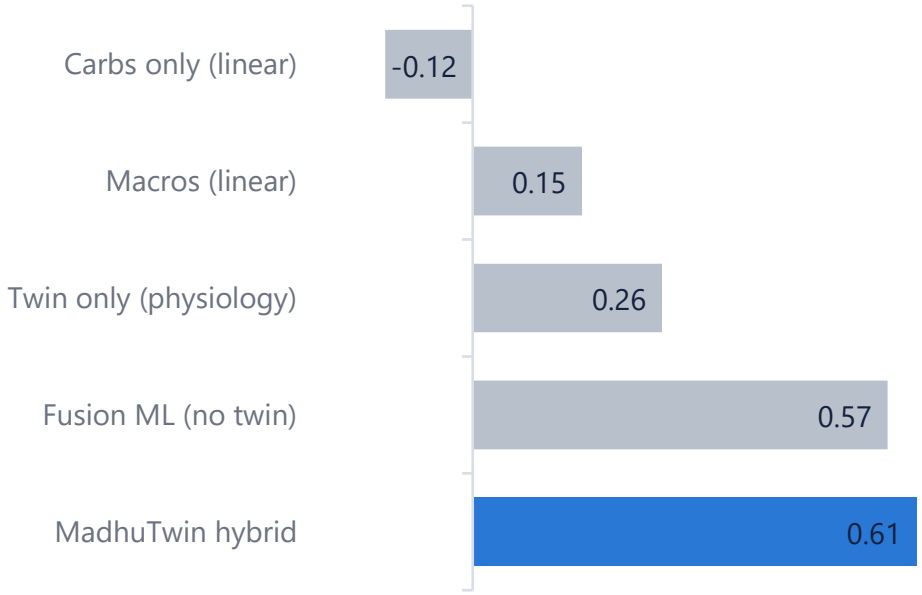
Where the physiological twin earns its place

Day-zero personalisation: twin two-hour RMSE vs days of onboarding data



21 % lower error on day zero. The record makes the twin usable before any sensor data; after about a day the streams dominate.

Meal response: predicted vs actual peak, Pearson r (664 meals)



Physiology beats carbohydrate counting. Carbs alone do not predict a person's response; the twin does. On the synthetic Indian cohort, 57 % of lows are flagged at least 10 minutes ahead.

What did not work, and what we did about it

Twin features vs the best fusion model

On rolling real-data forecasts, adding twin features to the ML layer did not beat the best fusion model. The twin's measured value is simulation, day-zero personalisation and meal-response prediction.

Three ideas rejected after validation

A disturbance observer, an activity-persistence prior and a fasting steady-state constraint were tested on held-out data and switched off: none improved accuracy.

A small, non-Indian real cohort

45 US adults, about ten days each. Indian diets, drugs and demographics are tested on a synthetic cohort that is always labelled and never mixed with real evidence.

Lows: trust the risk, not the line

Lows are rare in real data and point forecasts regress towards typical levels before them. The calibrated hypoglycaemia classifier (AUROC 0.85 real, 0.99 synthetic) is the primary signal.

Built for Indian clinics: diets, drugs, data law and standards

CLINICAL AND CULTURAL CONTEXT

Indian food, Indian prescriptions

54 Indian dishes with glycaemic index (IFCT 2017)

Teneligliptin, voglibose, glimepiride, gliclazide

Asian-Indian BMI and waist cut-offs; IDRS

Patient language recorded for notices and nudges

PRIVACY BY DESIGN

Aligned with the DPDP Act 2023

Purpose-bound consent per patient

Data minimisation and retention limits

Only open de-identified or fictional data

Decision support: no dosing instructions

INTEROPERABLE AND OPEN

Ready for ABDM-style exchange

HL7 FHIR R4 bundles

LOINC, SNOMED CT, ICD-10, WHO ATC

HL7 CGM summary codes; AGP report

Open source: Python, FastAPI, React

From prototype to clinic

NOW

Validated prototype

Real-data evidence, Indian synthetic cohort, clinician dashboard, FHIR export

NEXT SIX MONTHS

Clinical pilot

A remote-monitoring clinic in Bengaluru: alert-guided vs usual care on time in range and lows

TWELVE MONTHS

Scale

Photo-based Indian meal logging, lower-cost sensors, an on-device twin for low connectivity

BEYOND

More conditions

The same twin framework for hypertension and cardiovascular risk; CDSCO SaMD pathway

Who benefits. Endocrinologists and physicians running remote-monitoring clinics, diabetes educators, and community programmes where every specialist minute counts.

MADHUTWIN

Warn early. Explain why. Test before acting.

Anish Karthik Ayyagari · Indian Institute of Technology Kharagpur

